

On the Origin of Stochastic Photometric Variability in Massive Stars

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Submitted to ApJ

ABSTRACT

Recent high-precision photometric observations have revealed a new ubiquitous phenomena in early type stars: stochastic low-frequency photometric variability. There are currently two main ideas for the origin of this photometric variability: Internal Gravity Waves (IGW) launched by the convective core, or granulation/IGW produced by (sub)surface convection. Thanks to high resolution ground-based spectroscopy, many of the targets observed with K2 and TESS now have also good stellar parameters, including mass, luminosity and a measure of surface velocity fluctuations (macroturbulence). Here we show theoretical predictions for the trend of characteristic frequency and amplitude of the perturbation induced by subsurface convection in massive stars, and compare these predictions to observations. We found that both the characteristic frequency ν_{char} and amplitude α_0 of the observed photometric variability correlate very well with properties in the iron subsurface convection zone (FeCZ). While the amplitude of photometric variability also correlates with theoretical predictions of the flux of IGWs generated by core convection, the observed trend of characteristic frequency is at odds with theoretical expectations from convective turnover timescales in convective cores. This tentatively breaks the degeneracy between the two plausible models in favor of subsurface convection. We suggest a unified picture in which low-frequency photometric variability, surface turbulence, magnetic spots and wind variability in massive stars could all have the same origin: the presence of subsurface convection zones.

1. INTRODUCTION

Space Photometry revolution. New type of variability. Other mysteries at the surface of massive stars (turbulence, wind variability, spots). Stellar structure: convection in the core and convection in the envelope. IGWs and granulation. We discuss the properties of subsurface convection, in particular the maximum amplitude of waves that can be excited by their turbulent activity and their frequency as a function of the stellar temperature and luminosity. We show that the predicted trends of IGWs excited by turbulent convection in the FeCZ match very well the observations. We then proceed showing that while the trend with amplitude of the photometric variability could be reproduced by IGWs launched by core convection, the trend with excitation frequency in the core is opposite to the one observed at the surface. We discuss if wave propagation through the stellar envelope can in principle revert this trend,

and conclude this is unlikely. We conclude suggesting that subsurface convection represents a possible unifying mechanism for low-frequency photometric variability, surface turbulence, magnetic spots and wind variability in massive stars

This shows up as stochastic low-frequency variability that was observed in a large fraction of massive OB stars (see e.g. Balona 1992; Buysschaert et al. 2015; Bowman et al. 2019b).

2. METHODS

We calculated stellar evolution models using the Modules for Experiments in Stellar Astrophysics (MESA Paxton et al. 2011, 2013, 2015, 2018, 2019) software instrument. Details on the microphysics inputs to this software instrument are given in Appendix A. The inlists used to run our models can be found in this [GitHub repository](https://github.com/matteocantiello/rednoise)¹. We focused on the structure and convective properties of our models, in order to calculate typical frequencies and convective fluxes. Convection

is calculated in the framework of the Mixing Length Theory (Böhm-Vitense 1958, MLT hereafter), and we adopted $\alpha_{\text{MLT}} = 1.6$. While in efficient convection zones (e.g. stellar cores) the choice of this parameter does not change results much, this is not true in superadiabatic, inefficient convective regions close to the stellar surface (Jiang et al. 2015; Cantiello & Braithwaite 2019). We discuss in Section 4 how uncertainties in the choice of the α_{MLT} parameter affect our results.

Since we are interested in the excitation of surface phenomena, we focus on the upper part of convective zones and following Cantiello et al. (2009) we define the average of a quantity $\bar{q}$ as

$$\bar{q} \equiv \frac{1}{\alpha_{\text{MLT}} H_{\text{P}}} \int_{R_{\text{c}} - \alpha_{\text{MLT}} H_{\text{P}}}^{R_{\text{c}}} q(r) dr \quad (1)$$

where H_{P} is the pressure scale height calculated at the upper boundary (R_{c}) of the convective zone of interest. We tested using different average procedures, and found that our results do not depend much on the specific choice.

Using this definition, we extracted the average Mach number $\bar{M}_{\text{c}}$, convective velocity $\bar{v}_{\text{c}}$ and density $\bar{\rho}_{\text{c}}$ in the convective core and in the subsurface convection zones of our models. The most important subsurface convection zone for the massive stars discussed in this work is the Iron Convection Zone (FeCZ) (Cantiello et al. 2009), although in the low luminosity regime He-driven convection zones could play a role as well (Cantiello & Braithwaite 2019). We define the characteristic frequency as

$$\nu_{\text{char}} \equiv \frac{1}{2\pi\tau}, \quad (2)$$

where τ is a characteristic timescale. When calculating the characteristic frequency in our models, we set τ to be the average convective turnover time

$$\tau_{\text{c}} = \alpha_{\text{MLT}} \bar{H}_{\text{P}} / \bar{v}_{\text{c}} \quad (3)$$

calculated either in the FeCZ or in the convective core. We proceeded to calculate convective turnover timescales and other relevant properties like convective fluxes $F_{\text{c}} = \bar{\rho}_{\text{c}} \bar{v}_{\text{c}}^3$. We did this as a function of both mass and evolutionary history for stars ranging from $5 - 120 M_{\odot}$. Here we present results for an initial metallicity of $Z=0.02$, but in Appendix C we report results for model grids with $Z = 0.014, 0.006$ and 0.002 as well.

We compare these quantities with the observed characteristic frequency ν_{char} and amplitude α_0 of stochastic photometric variability. These quantities are derived by fitting the stochastically-excited low-frequency power

excess in a power density spectrum using a Lorentzian function (e.g. Bowman et al. 2019a):

$$\alpha(\nu) = \frac{\alpha_0}{1 + \left(\frac{\nu}{\nu_{\text{char}}}\right)^{\gamma}} + P_{\text{W}}. \quad (4)$$

This shows that α_0 represents the amplitude at zero frequency. ν_{char} is defined as in Eq. 2 with τ the typical timescale of the stochastic photometric variability. Finally, γ is the gradient of the linear part of the profile and P_{W} is a white noise term.

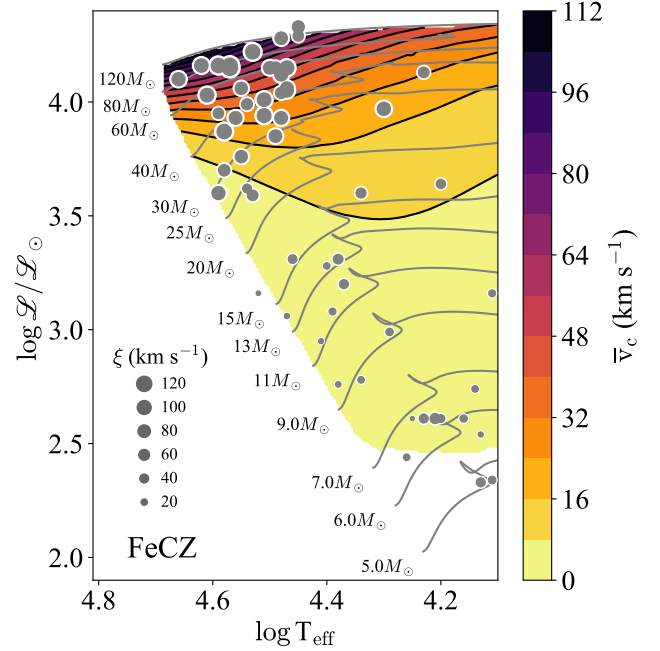


Figure 1. Average convective velocities in the FeCZ as function of the location of stellar models in the spectroscopic H-R Diagram. We also show the observed stars with detected macroturbulent velocity as grey dots. The size of the dot is proportional to the observed macroturbulence.

3. RESULTS

We want to test a possible correlation between the properties of the FeCZ and observed photospheric phenomena, in particular photometric variability and turbulent velocities at the stellar surface (macroturbulence). If photometric variability was caused by the FeCZ, a natural choice of proxy for the typical timescale of this variability is the convective turnover timescale. We found typical values of the convective turnover timescale to be about $\sim 0.1 \dots 2$ d in the FeCZ, with a tendency for shorter values in models with high effective temperature and surface luminosity (Cantiello et al. 2009). To quantitatively compare to the characteristic

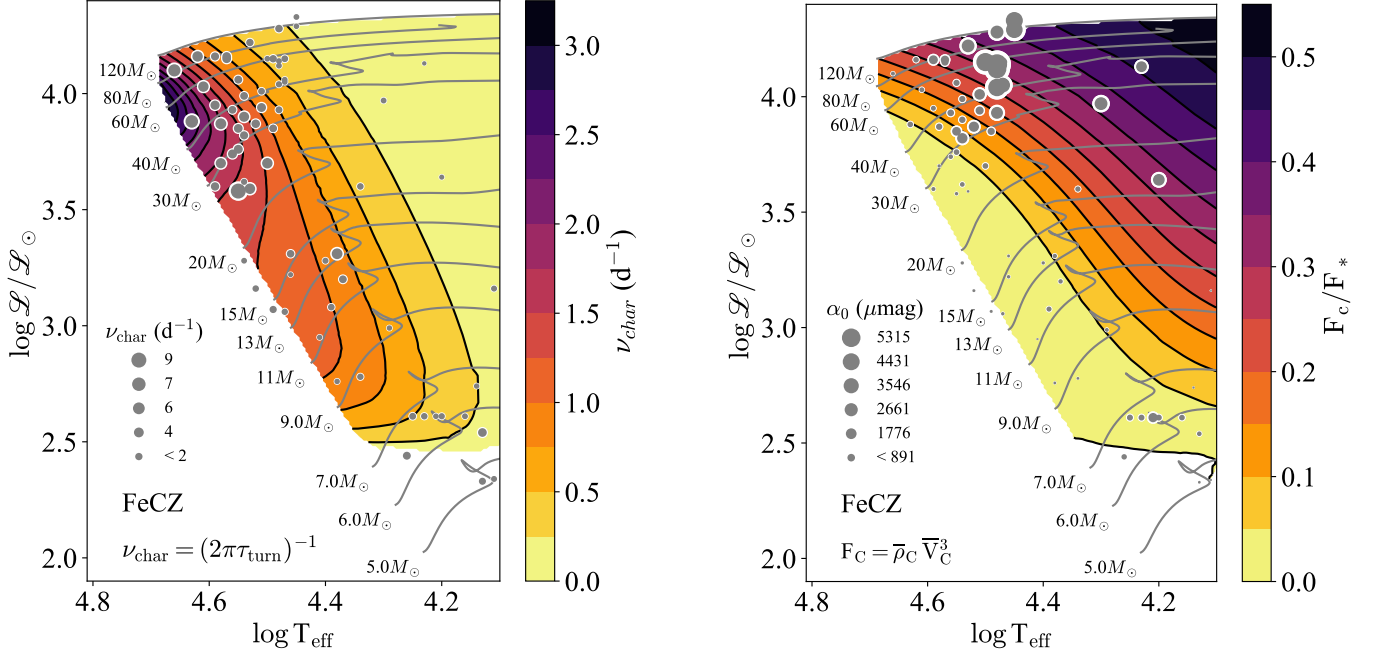


Figure 2. Left panel: Characteristic frequency ν_{char} in the FeCZ as a function of the location of stellar models in the spectroscopic H-R Diagram. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} . Right panel: Ratio of FeCZ convective flux to the total stellar flux in the spectroscopic H-R Diagram. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

frequencies of the observed stochastic photometric variability in massive stars, we proceed mapping our theoretical predictions for the quantity ν_{char} (eq. 2) on a spectroscopic H-R diagram (Langer & Kudritzki 2014). This reveals a very good agreement, with our models reproducing both the typical values of the observed ν_{char} and the trend with $\log L$ and $\log T_{\text{eff}}$ (Fig. 2). While the characteristic frequencies found by Bowman et al. (2020) seem to be larger by a factor of ≈ 3 , our predictions for the turnover timescale are affected by uncertainty in the convective velocities arising from the MLT treatment ($v_c \propto \alpha_{\text{MLT}}^3$, so $\nu_{\text{char}} \propto \alpha_{\text{MLT}}^2$). Since α_{MLT} is uncertain by a factor of 2 or so, our estimates of the characteristic frequency in the FeCZ are uncertain by a factor of ≈ 4 .

We proceed with mapping quantities that measure the amplitude of perturbations at the location of the FeCZ. We then check if some of these properties correlate with the amplitude of observed photospheric phenomena, including turbulent velocity fluctuations and photometric variability. While here we are agnostic about the particular mechanism, we assume that some physical process connects the subsurface layers to the photosphere. We discuss the most likely candidates for this process in Sec. 4.1. We show the average convective velocity in the FeCZ in Fig. 1. Velocities of the order $10 \dots 100 \text{ km s}^{-1}$

are found across the upper H-R diagram, with a trend of increasing v_c for higher luminosities. This trend is similar to what is observed for the macroturbulent velocity field at the surface of massive stars. This is in agreement with the results of Grassitelli et al. (2015), who found a tight correlation between the turbulent pressure ($P_{\text{turb}} \propto v_c^2$) in the FeCZ and spectroscopically-derived macroturbulent velocities in massive stars. Interestingly, the range of the velocity field in the FeCZ is also similar to the observed range of macroturbulent velocities. Again the caveat is the large uncertainty (a factor ≈ 10) of the convective velocities calculated using the MLT in inefficient convective zones (Cantiello & Braithwaite 2019). We plot the average Mach number $\bar{M}_c$ of convection in the FeCZ in Fig. 6, showing that contrary to core convection the FeCZ is just mildly subsonic, with values of the Mach number ranging from 0.01 to 0.3. The relative convective flux in these regions is shown in Fig. 2, where we also report the the observed amplitude of photometric variability. Values of F_c/F_* tend to increase with increasing $\log L$ and decreasing $\log T_{\text{eff}}$, a trend that is also found for the values of observed photometric variability.

3.1. Core Convection and Photometric Variability

Early-type stars possess convective cores during their main sequence. Turbulent convection in these convective cores excites internal gravity waves (Goldreich & Kumar 1990; Rogers et al. 2013; Lecoanet & Quataert 2013; Shiode et al. 2013; Edelmann et al. 2019), that can propagate through the stellar envelope and reach the stellar surface (e.g. Lecoanet et al. 2019; Ratnasingham et al. 2020). These waves could in principle be responsible for both the observed macroturbulence and photometric variability in early-type stars (Aerts et al. 2009; Simón-Díaz et al. 2017; Bowman et al. 2019a,b). However, Lecoanet et al. (2019) studied the effects of wave propagation and radiative damping in the stellar envelope, concluding that it is unlikely that the observed photometric variability is caused by internal gravity waves launched by the convective core. One of the biggest uncertainties is the excitation spectrum of internal gravity waves. While Coustou et al. (2018) found... Edelmann et al. (2019) claims that...

Here we focus on the core convective properties of our models, and similarly to the case of the FeCZ we look for correlations with the amplitude and characteristic frequency of the observed surface variability. The flux of Internal Gravity Waves excited by turbulent convection at the core boundary is $F_{\text{IGW}} = F_* \bar{M}_c$ (Goldreich & Kumar 1990). We expect the relative surface radial displacement ξ_r/R produced by internal gravity waves launched from the core to be proportional to the quantity $\sqrt{F_{\text{IGW}}/F_*} = \sqrt{\bar{M}_c}$ (see Appendix ??). We show this quantity in Fig. 3. The Mach number in the convective core is on the order of $10^{-3} \dots 10^{-4}$, with a trend of increasing values for lower $\log T_{\text{eff}}$ and higher $\log \mathcal{L}$. This trend matches the observed trend in the photometric variability pretty well.

Next, we focus on the characteristic timescale of waves excited by core convection. We expect the maximum of the IGW flux to be launched at frequencies close to $\nu_{\text{char}} = (2\pi\tau_c)^{-1}$, with τ_c defined in Eq. 3. Fig. 3 shows that the predicted values of ν_{char} are in the range $0.02 \dots 0.008 \text{ d}^{-1}$ ($0.2 \dots 0.08 \mu\text{Hz}$), in agreement with the results of Shiode et al. (2013). These values are about 2 orders of magnitude smaller than the typical characteristic frequencies of stochastic photometric variability observed by (Bowman et al. 2019a,b). Wave propagation through the stellar envelope affects the spectrum, changing the frequency of maximum power of waves reaching the surface (Lecoanet et al. 2019). Radiative damping in particular can efficiently suppress waves at low frequency, so that the peak of the spectrum observed at the surface is expected to move to higher frequencies. The amplitude of this effect depends on the envelope properties, which are a function

of $\log T_{\text{eff}}$ and $\log \mathcal{L}$. However, we expect the trends of predicted ν_{char} as function of $\log T_{\text{eff}}$ and $\log \mathcal{L}$ to be mostly determined by the core convective properties (Daniel check / expand on this?). In Fig. 3 the prediction is that ν_{char} should increase with both $\log T_{\text{eff}}$ and $\log \mathcal{L}$, so that the characteristic frequencies of peak IGWs should increase as stars evolve on the main sequence. This is exactly the opposite of what is observed: the characteristic frequencies of stochastic photometric variability is largest for stars early on on the main sequence, and seem to decrease as stars evolve. Therefore the data seems to break the degeneracy between the proposed models in favor of subsurface convection.

3.2. Macroturbulence

The hypothesis that subsurface convection is responsible for surface turbulence is corroborated by the match between trends in the FeCZ properties (i.e. convective velocities and turbulent pressure) and the observed micro and macroturbulence (Cantiello et al. 2009; Grassitelli et al. 2015). In particular, there appears to be a critical magnetic field above which macroturbulence disappears, and this critical field strength corresponds closely to the critical field needed to stabilize the FeCZ (MacDonald & Petit 2019; Sundqvist et al. 2013; Jermyn & Cantiello 2020).

We can analogously define a critical magnetic field strength B_{crit} which suffices to reflect IGWs before they reach the photosphere. Using the dispersion relation for IGWs in a magnetic medium, the radial component of this field is (Fuller et al. 2015)

$$B_{r,\text{crit}} = \frac{\omega}{2k_r} \sqrt{4\pi\rho}, \quad (5)$$

where $\omega = \nu/2\pi$ is the angular frequency of the waves, ρ is the density, and k_r is the radial wavenumber. For IGWs the radial wave-number is related to the azimuthal quantum number ℓ by

$$k_r \approx \frac{\sqrt{\ell(\ell+1)}}{r} \left(\frac{N}{\omega} \right), \quad (6)$$

where N is the Brünt-Väisälä frequency. So

$$B_{r,\text{crit}} \approx \frac{\omega^2 r}{N} \sqrt{\frac{4\pi\rho}{2\ell(\ell+1)}} \quad (7)$$

$$\approx \frac{\omega^2 r}{\ell N} \sqrt{4\pi\rho}. \quad (8)$$

Note that because this decreases with increasing ℓ , all waves of a given frequency are reflected if that with $\ell = 1$ is reflected.

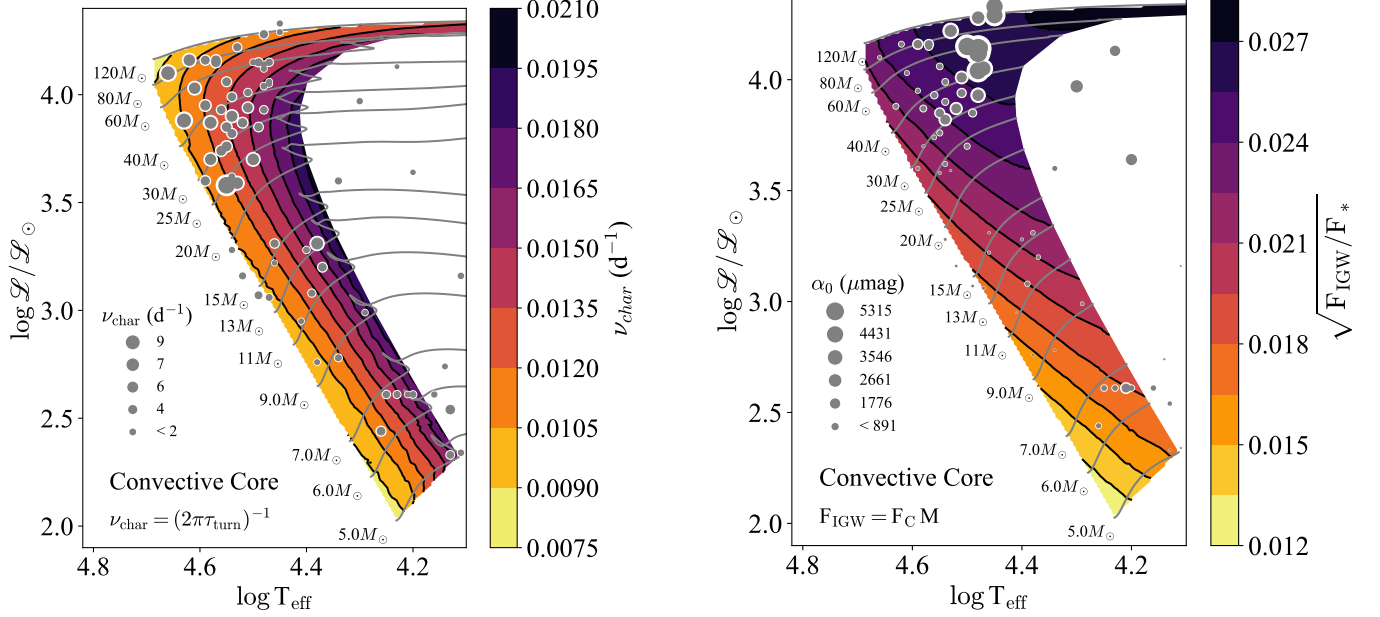


Figure 3. Left panel: Characteristic frequency ν_{char} in the convective core as function of the location of stellar models in the spectroscopic HRD. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} . Right panel: Square root of the ratio between the flux of gravity waves launched by the convective core and the total stellar flux. The IGW flux is calculated multiplying the core convective flux by the average convective Mach number (in the top pressure scale height of the convective region). We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

We computed $B_{r,\text{crit}}$ for several values of ℓ and ν for a main-sequence model of a $30M_{\odot}$ star as a function of radius, shown in Fig. 3.2. For frequencies comparable to those of core convection the critical magnetic field strength is of order 10^{-2} G to 10^{-1} G. Strong macroturbulence is observed in similar-mass O-type stars with magnetic fields up to 2.5 kG (e.g. HD 191612) (Sundqvist et al. 2013), so macroturbulence in those stars is unlikely to be due to IGW coming from their cores if those IGW have similar frequencies to that of core convection.

For frequencies comparable to the observed ν_{char} , on the order of 3 d^{-1} , the critical magnetic field is much larger, on the order of 300 G to 1 kG. This makes an explanation of macroturbulence based on IGW marginally inconsistent with observations showing strong macroturbulence up to field strengths of 2.5 kG. However, because equation (7) is a strong function of ω , and hence of ν , it is possible that these strongly magnetized stars just have larger ν_{char} . For instance if HD 191612 has $\nu_{\text{char}} \approx 5 \text{ d}^{-1}$ then it would have $B_{r,\text{crit}} \approx 3 \text{ kG}$ for the $\ell = 1$ mode.

A further prediction of this calculation is that, if IGW are the cause of macroturbulence in these stars, we should expect the strength of macroturbulence to decline with increasing magnetic field strength as more

and more modes are reflected before they reach the surface. Sundqvist et al. (2013) find no such trend, and the only strong correlation they see of macroturbulence with magnetic field strength is that it vanishes when the magnetic field exceeds the FeCZ shutoff strength. This again points against an explanation of macroturbulence based on IGWs.

4. DISCUSSION

4.1. Above the Iron Convection Zone

The presence of a subsurface convection zone can result in variability of photospheric properties via a number of processes, including wave excitation (Cantiello et al. 2009; Grassitelli et al. 2015) and magnetic buoyancy (Cantiello & Braithwaite 2011). Moreover, multi dimensional simulations including radiation have been performed in a restricted range of the parameter space, and show that the full turbulent manifestation of these convective regions extends up to the stellar surface (Jiang et al. 2015, 2017, 2018). Therefore it could be that the observed stochastic photometric variability represents the direct manifestation of turbulent, radiation-dominated convection at the stellar surface. The simplified MLT treatment in our one dimensional calculations is unable to capture the complex phenomenology of these layers, but the fact that it can reproduce both the

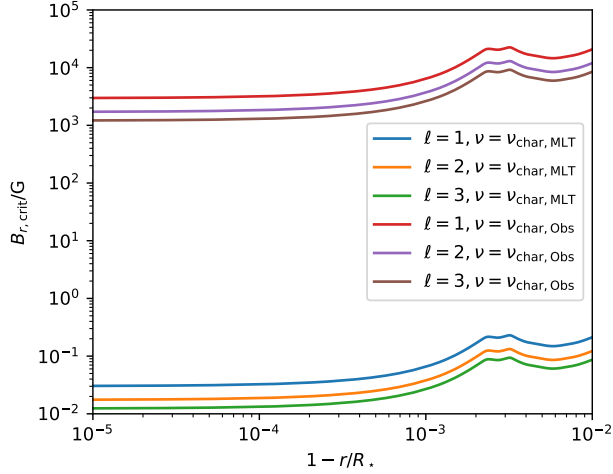


Figure 4. The critical radial magnetic field strength needed to reflect waves of different ℓ and ν is shown as a function of fractional depth for a $30M_{\odot}$ stellar model at an age of 4 Myr. The frequency $\nu_{\text{char,MLT}}$ is that given by equation (2) but evaluated for the core convection zone instead of the subsurface FeCZ. The frequency $\nu_{\text{char,Obs}}$ is a typical value of 3 d^{-1} for the observations.

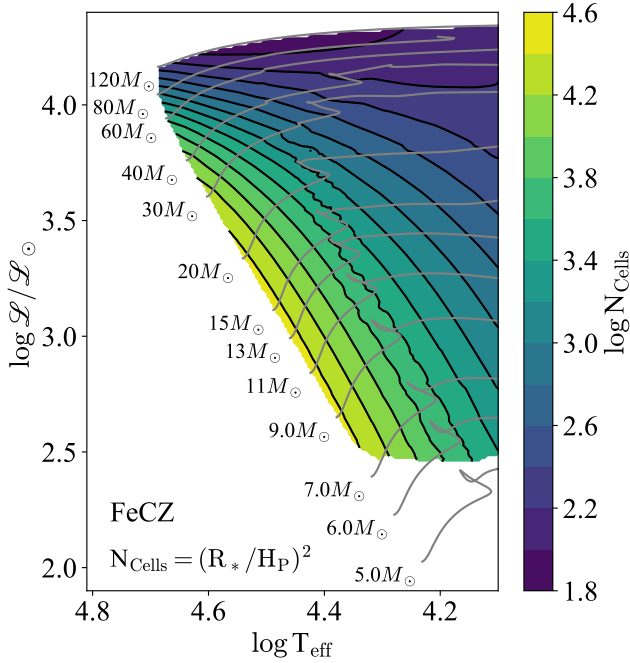


Figure 5. Expected number of convective cells in the FeCZ. This is calculated assuming the size of convective cells is about the average pressure scale height in the FeCZ

timescales and the trends in amplitude of the observed photometric variability is compelling. It calls for radiation hydrodynamics simulations covering the parameter

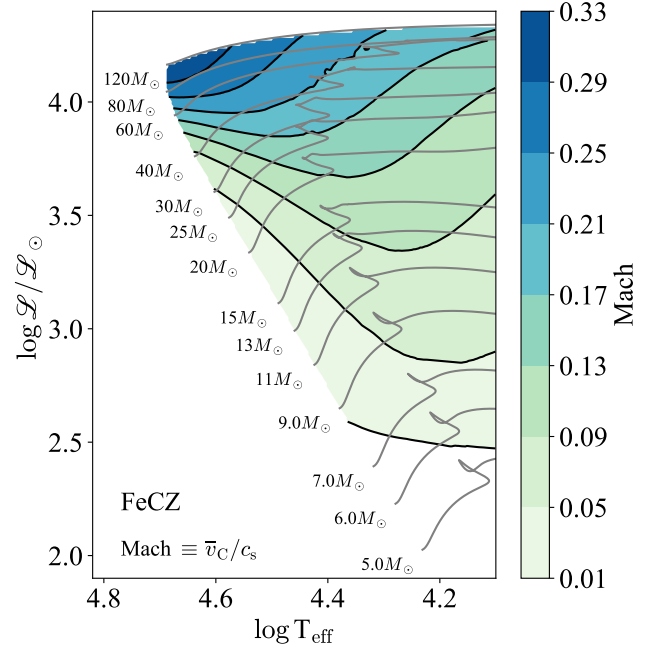


Figure 6. Mach number in the FeCZ.

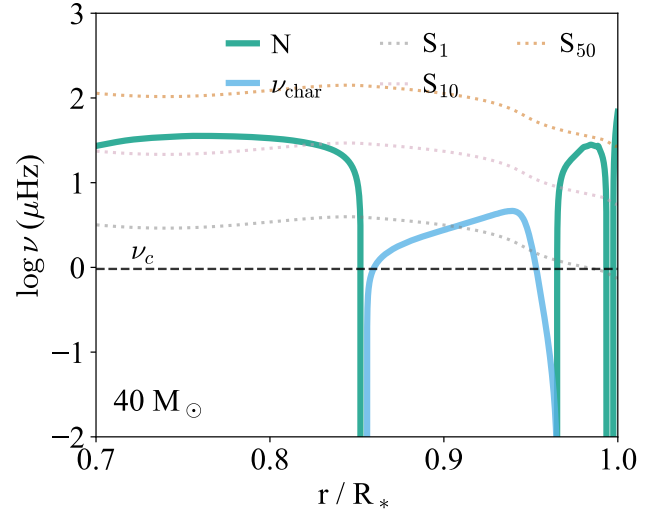


Figure 7. Propagation diagram for a $40M_{\odot}$ model showing the outer 30% of the star in radial coordinate.

space of the TESS observations in order to unravel the precise mechanism connecting (sub)surface convection zones to the observed surface variability.

4.2. A unified model for surface phenomena in massive stars

The presence of subsurface convection can simultaneously account for a large variety of puzzling phenomena that appear ubiquitous at the surface of early-type

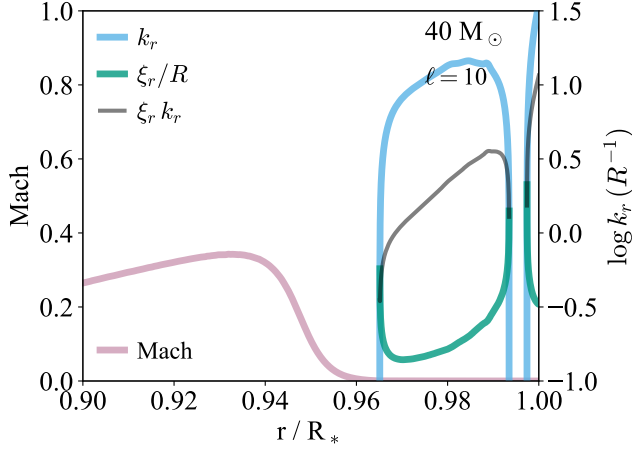


Figure 8. Properties of waves launched by the FeCZ in a $40M_{\odot}$ model. The plot shows the outer 10% of the star in radial coordinate.

stars. First we have microturbulence and macroturbulence, and both can be accounted for by velocity fields excited by the underlying FeCZ subsurface convection zone (Cantiello et al. 2009; Grassitelli et al. 2015). The only stars with no macroturbulence appear to have magnetic fields strong enough to shut off convection in the FeCZ, while stars with slightly lower surface magnetic fields show normal values of macroturbulence. Then we have the observation of bright spots in early-type stars (e.g. Ramiamananantsoa et al. 2014), which can be explained with the presence of magnetic spots rising from subsurface convective layers (Cantiello & Braithwaite 2011, 2019). These photospheric magnetic spots can then be responsible for another ubiquitous phenomena in early-type stars, the presence of discrete absorption components (DACs) in their UV spectra (e.g. Howarth & Prinja 1989; Cranmer & Owocki 1996; Fullerton et al. 1997; Kaper et al. 1997). Overall (Sub)surface convection is expected to drive velocity and density perturbations at the stellar surface (e.g. Cantiello et al. 2009). These perturbations could explain the ubiquitous line profile variability (LPV, Fullerton et al. 1996) and provide a seed for the rapid development of wind instabilities that lead to wind clumping (e.g. Owocki et al. 1988; Sundqvist & Owocki 2013). Finally, these surface velocity and density perturbations would necessarily induce photometric variability, and we have shown here that the observed low-frequency stochastic noise detected in

the majority of massive stars correlates very well with the presence and properties of the FeCZ.

An economical hypothesis emerges: the presence of (sub)surface convection, and in particular of the FeCZ, could be the common underlying physical cause for the appearance of turbulence, magnetic spots, low-frequency photometric variability, as well as wind and spectroscopic variability in early-type stars.

5. CONCLUSIONS

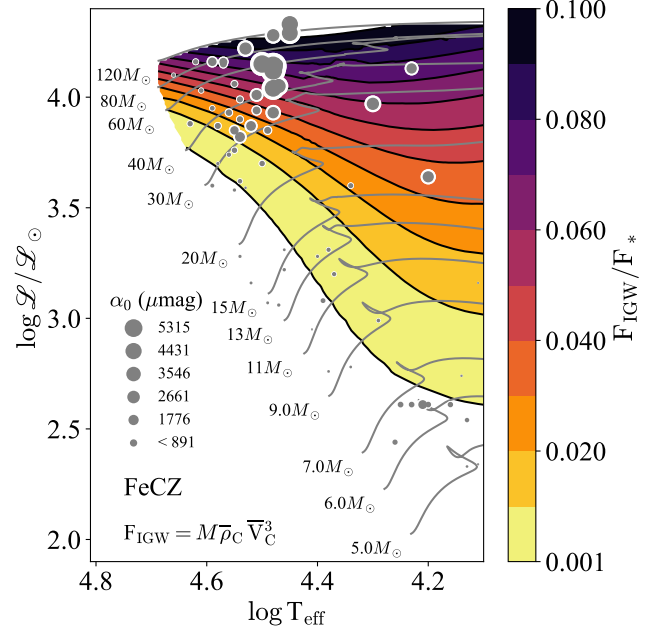


Figure 9. Flux of gravity waves launched by the FeCZ. This is calculated multiplying the convective flux in the FeCZ by the average convective Mach number (this is calculated in the top pressure scale height of the convective region). We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

ACKNOWLEDGMENTS

The Center for Computational Astrophysics at the Flatiron Institute is supported by the Simons Foundation. This research was supported in part by the National Science Foundation under Grant No. NSF PHY-1748958 and by the Gordon and Betty Moore Foundation through Grant GBMF7392

APPENDIX

A. SOFTWARE DETAILS

Calculations were done with MESA version 11701. The MESA EOS is a blend of the OPAL Rogers & Nayfonov (2002), SCVH Saumon et al. (1995), PTEH Pols et al. (1995), HELM Timmes & Swesty (2000), and PC Potekhin & Chabrier (2010) EOSes.

Radiative opacities are primarily from OPAL (Iglesias & Rogers 1993, 1996), with low-temperature data from Ferguson et al. (2005) and the high-temperature, Compton-scattering dominated regime by Buchler & Yueh (1976). Electron conduction opacities are from Cassisi et al. (2007).

Nuclear reaction rates are a combination of rates from NACRE (Angulo et al. 1999), JINA REACLIB (Cyburt et al. 2010), plus additional tabulated weak reaction rates Fuller et al. (1985); Oda et al. (1994); Langanke & Martínez-Pinedo (2000). Screening is included via the prescription of Chugunov et al. (2007). Thermal neutrino loss rates are from Itoh et al. (1996).

The inlists, processing scripts, and model output are available at Zenodo.org.

B. RADIAL DISPLACEMENT FROM IGWS

C. GRIDS AT DIFFERENT METALLICITIES

Here we present results for the critical and surface magnetic field at metallicities of $Z=0.014$, $Z=0.006$, and $Z=0.002$, representing early-type stars in the Galaxy (MW), Large Magellanic Cloud (LMC), and Small Magellanic Cloud (SMC) respectively (e.g. Yusof et al. 2013). The initial Helium content of the grids is $Y=0.2659$ (MW), $Y=0.2559$ (LMC), and $Y=0.2508$ (SMC). The metallicity is initialized scaling the standard solar composition of Grevesse & Sauval (1998).

REFERENCES

- Aerts, C., Puls, J., Godart, M., & Dupret, M. A. 2009, *A&A*, 508, 409, doi: [10.1051/0004-6361/200810471](https://doi.org/10.1051/0004-6361/200810471)
- Angulo, C., Arnould, M., Rayet, M., et al. 1999, *Nuclear Physics A*, 656, 3, doi: [10.1016/S0375-9474\(99\)00030-5](https://doi.org/10.1016/S0375-9474(99)00030-5)
- Böhm-Vitense, E. 1958, *ZA*, 46, 108
- Bowman, D. M., Burssens, S., Simón-Díaz, S., et al. 2020, *A&A*, 640, A36, doi: [10.1051/0004-6361/202038224](https://doi.org/10.1051/0004-6361/202038224)
- Bowman, D. M., Aerts, C., Johnston, C., et al. 2019a, *A&A*, 621, A135, doi: [10.1051/0004-6361/201833662](https://doi.org/10.1051/0004-6361/201833662)
- Bowman, D. M., Burssens, S., Pedersen, M. G., et al. 2019b, *Nature Astronomy*, 3, 760, doi: [10.1038/s41550-019-0768-1](https://doi.org/10.1038/s41550-019-0768-1)
- Buchler, J. R., & Yueh, W. R. 1976, *ApJ*, 210, 440, doi: [10.1086/154847](https://doi.org/10.1086/154847)
- Cantiello, M., & Braithwaite, J. 2011, *A&A*, 534, A140, doi: [10.1051/0004-6361/201117512](https://doi.org/10.1051/0004-6361/201117512)
- . 2019, *ApJ*, 883, 106, doi: [10.3847/1538-4357/ab3924](https://doi.org/10.3847/1538-4357/ab3924)
- Cantiello, M., Langer, N., Brott, I., et al. 2009, *A&A*, 499, 279, doi: [10.1051/0004-6361/200911643](https://doi.org/10.1051/0004-6361/200911643)
- Cassisi, S., Potekhin, A. Y., Pietrinferni, A., Catelan, M., & Salaris, M. 2007, *ApJ*, 661, 1094, doi: [10.1086/516819](https://doi.org/10.1086/516819)
- Chugunov, A. I., Dewitt, H. E., & Yakovlev, D. G. 2007, *PhRvD*, 76, 025028, doi: [10.1103/PhysRevD.76.025028](https://doi.org/10.1103/PhysRevD.76.025028)
- Couston, L.-A., Lecoanet, D., Favier, B., & Le Bars, M. 2018, *Journal of Fluid Mechanics*, 854, R3, doi: [10.1017/jfm.2018.669](https://doi.org/10.1017/jfm.2018.669)
- Cranmer, S. R., & Owocki, S. P. 1996, *ApJ*, 462, 469, doi: [10.1086/177166](https://doi.org/10.1086/177166)
- Cyburt, R. H., Amthor, A. M., Ferguson, R., et al. 2010, *ApJS*, 189, 240, doi: [10.1088/0067-0049/189/1/240](https://doi.org/10.1088/0067-0049/189/1/240)
- Edelmann, P. V. F., Ratnasingam, R. P., Pedersen, M. G., et al. 2019, *ApJ*, 876, 4, doi: [10.3847/1538-4357/ab12df](https://doi.org/10.3847/1538-4357/ab12df)
- Ferguson, J. W., Alexander, D. R., Allard, F., et al. 2005, *ApJ*, 623, 585, doi: [10.1086/428642](https://doi.org/10.1086/428642)
- Fuller, G. M., Fowler, W. A., & Newman, M. J. 1985, *ApJ*, 293, 1, doi: [10.1086/163208](https://doi.org/10.1086/163208)
- Fuller, J., Cantiello, M., Stello, D., Garcia, R. A., & Bildsten, L. 2015, *Science*, 350, 423, doi: [10.1126/science.aac6933](https://doi.org/10.1126/science.aac6933)
- Fullerton, A. W., Gies, D. R., & Bolton, C. T. 1996, *ApJS*, 103, 475, doi: [10.1086/192285](https://doi.org/10.1086/192285)
- Fullerton, A. W., Massa, D. L., Prinja, R. K., Owocki, S. P., & Cranmer, S. R. 1997, *A&A*, 327, 699
- Goldreich, P., & Kumar, P. 1990, *ApJ*, 363, 694, doi: [10.1086/169376](https://doi.org/10.1086/169376)
- Grassitelli, L., Fossati, L., Simón-Díaz, S., et al. 2015, *ApJL*, 808, L31, doi: [10.1088/2041-8205/808/1/L31](https://doi.org/10.1088/2041-8205/808/1/L31)
- Grevesse, N., & Sauval, A. J. 1998, *SSRv*, 85, 161, doi: [10.1023/A:1005161325181](https://doi.org/10.1023/A:1005161325181)
- Howarth, I. D., & Prinja, R. K. 1989, *ApJS*, 69, 527, doi: [10.1086/191321](https://doi.org/10.1086/191321)
- Iglesias, C. A., & Rogers, F. J. 1993, *ApJ*, 412, 752, doi: [10.1086/172958](https://doi.org/10.1086/172958)
- . 1996, *ApJ*, 464, 943, doi: [10.1086/177381](https://doi.org/10.1086/177381)
- Itoh, N., Hayashi, H., Nishikawa, A., & Kohyama, Y. 1996, *ApJS*, 102, 411, doi: [10.1086/192264](https://doi.org/10.1086/192264)

- Jermyn, A. S., & Cantiello, M. 2020, *ApJ*, 900, 113, doi: [10.3847/1538-4357/ab9e70](https://doi.org/10.3847/1538-4357/ab9e70)
- Jiang, Y.-F., Cantiello, M., Bildsten, L., Quataert, E., & Blaes, O. 2015, *ApJ*, 813, 74, doi: [10.1088/0004-637X/813/1/74](https://doi.org/10.1088/0004-637X/813/1/74)
- . 2017, *ApJ*, 843, 68, doi: [10.3847/1538-4357/aa77b0](https://doi.org/10.3847/1538-4357/aa77b0)
- Jiang, Y.-F., Cantiello, M., Bildsten, L., et al. 2018, *Nature*, 561, 498, doi: [10.1038/s41586-018-0525-0](https://doi.org/10.1038/s41586-018-0525-0)
- Kaper, L., Henrichs, H. F., Fullerton, A. W., et al. 1997, *A&A*, 327, 281
- Langanke, K., & Martínez-Pinedo, G. 2000, *Nuclear Physics A*, 673, 481, doi: [10.1016/S0375-9474\(00\)00131-7](https://doi.org/10.1016/S0375-9474(00)00131-7)
- Langer, N., & Kudritzki, R. P. 2014, *A&A*, 564, A52, doi: [10.1051/0004-6361/201423374](https://doi.org/10.1051/0004-6361/201423374)
- Lecoanet, D., & Quataert, E. 2013, *MNRAS*, 430, 2363, doi: [10.1093/mnras/stt055](https://doi.org/10.1093/mnras/stt055)
- Lecoanet, D., Cantiello, M., Quataert, E., et al. 2019, *ApJL*, 886, L15, doi: [10.3847/2041-8213/ab5446](https://doi.org/10.3847/2041-8213/ab5446)
- MacDonald, J., & Petit, V. 2019, *MNRAS*, 487, 3904, doi: [10.1093/mnras/stz1545](https://doi.org/10.1093/mnras/stz1545)
- Oda, T., Hino, M., Muto, K., Takahara, M., & Sato, K. 1994, *Atomic Data and Nuclear Data Tables*, 56, 231, doi: [10.1006/adnd.1994.1007](https://doi.org/10.1006/adnd.1994.1007)
- Owocki, S. P., Castor, J. I., & Rybicki, G. B. 1988, *ApJ*, 335, 914, doi: [10.1086/166977](https://doi.org/10.1086/166977)
- Paxton, B., Bildsten, L., Dotter, A., et al. 2011, *ApJS*, 192, 3, doi: [10.1088/0067-0049/192/1/3](https://doi.org/10.1088/0067-0049/192/1/3)
- Paxton, B., Cantiello, M., Arras, P., et al. 2013, *ApJS*, 208, 4, doi: [10.1088/0067-0049/208/1/4](https://doi.org/10.1088/0067-0049/208/1/4)
- Paxton, B., Marchant, P., Schwab, J., et al. 2015, *ApJS*, 220, 15, doi: [10.1088/0067-0049/220/1/15](https://doi.org/10.1088/0067-0049/220/1/15)
- Paxton, B., Schwab, J., Bauer, E. B., et al. 2018, *ApJS*, 234, 34, doi: [10.3847/1538-4365/aaa5a8](https://doi.org/10.3847/1538-4365/aaa5a8)
- Paxton, B., Smolec, R., Schwab, J., et al. 2019, *ApJS*, 243, 10, doi: [10.3847/1538-4365/ab2241](https://doi.org/10.3847/1538-4365/ab2241)
- Pols, O. R., Tout, C. A., Eggleton, P. P., & Han, Z. 1995, *MNRAS*, 274, 964, doi: [10.1093/mnras/274.3.964](https://doi.org/10.1093/mnras/274.3.964)
- Potekhin, A. Y., & Chabrier, G. 2010, *Contributions to Plasma Physics*, 50, 82, doi: [10.1002/ctpp.201010017](https://doi.org/10.1002/ctpp.201010017)
- Ramaramananantsoa, T., Moffat, A. F. J., Chené, A.-N., et al. 2014, *MNRAS*, 441, 910, doi: [10.1093/mnras/stu619](https://doi.org/10.1093/mnras/stu619)
- Ratnasingam, R. P., Edelmann, P. V. F., & Rogers, T. M. 2020, *MNRAS*, doi: [10.1093/mnras/staa2296](https://doi.org/10.1093/mnras/staa2296)
- Rogers, F. J., & Nayfonov, A. 2002, *ApJ*, 576, 1064, doi: [10.1086/341894](https://doi.org/10.1086/341894)
- Rogers, T. M., Lin, D. N. C., McElwaine, J. N., & Lau, H. H. B. 2013, *ApJ*, 772, 21, doi: [10.1088/0004-637X/772/1/21](https://doi.org/10.1088/0004-637X/772/1/21)
- Saumon, D., Chabrier, G., & van Horn, H. M. 1995, *ApJS*, 99, 713, doi: [10.1086/192204](https://doi.org/10.1086/192204)
- Shiode, J. H., Quataert, E., Cantiello, M., & Bildsten, L. 2013, *MNRAS*, 430, 1736, doi: [10.1093/mnras/sts719](https://doi.org/10.1093/mnras/sts719)
- Simón-Díaz, S., Godart, M., Castro, N., et al. 2017, *A&A*, 597, A22, doi: [10.1051/0004-6361/201628541](https://doi.org/10.1051/0004-6361/201628541)
- Sundqvist, J. O., & Owocki, S. P. 2013, *MNRAS*, 428, 1837, doi: [10.1093/mnras/sts165](https://doi.org/10.1093/mnras/sts165)
- Sundqvist, J. O., Petit, V., Owocki, S. P., et al. 2013, *Monthly Notices of the Royal Astronomical Society*, 433, 2497, doi: [10.1093/mnras/stt921](https://doi.org/10.1093/mnras/stt921)
- Timmes, F. X., & Swesty, F. D. 2000, *ApJS*, 126, 501, doi: [10.1086/313304](https://doi.org/10.1086/313304)
- Yusof, N., Hirschi, R., Meynet, G., et al. 2013, *MNRAS*, 433, 1114, doi: [10.1093/mnras/stt794](https://doi.org/10.1093/mnras/stt794)